

Edge Negotiator: methodology and architectural rationale framework

§1 Scope and chapter role within the dissertation

This chapter serves as the methodological spine of the Edge Negotiator dissertation. It sits between the literature review (Chapter 2) and the implementation (Chapter 4), performing three functions: **documenting every locked architectural decision with its rationale and rejected alternatives**, establishing the evaluation framework that governs what counts as evidence in Chapters 5-6, and pre-empting viva challenges by making trade-offs, scope constraints, and epistemic limitations explicit rather than discoverable. The chapter does not relitigate decisions already made — it records them as outcomes with full provenance. Every requirement traces forward to a SUMO test; every claim carries a declared evidential grade. The structure follows Michael Nygard's Architecture Decision Record pattern, extended with threat modelling, requirements traceability, and an evidential standards framework flagged by Lee Stott as essential for methodological rigour.

§2 Six component sections

(A) Architectural Decision Records

The ADR format adopted follows Nygard's original lightweight structure (<https://www.cognitect.com/blog/2011/11/15/documenting-architecture-decisions> [archived: web.archive.org/web/2024/https://www.cognitect.com/blog/2011/11/15/documenting-architecture-decisions]), extended with an "Alternatives Considered" section from the MADR 4.0 template (<https://github.com/adr/madr> [archived: web.archive.org/web/2024/https://github.com/adr/madr]) and an "Evidence" section to satisfy Lee Stott's evidential standards requirement. ThoughtWorks placed ADRs in the "Adopt" ring of their Technology Radar (Red Hat Thoughtworks) (<https://www.thoughtworks.com/en-us/radar/techniques/lightweight-architecture-decision-records> [archived: web.archive.org/web/2024/https://www.thoughtworks.com/en-us/radar/techniques/lightweight-architecture-decision-records]).

ADR template (all seven decisions use this structure):

ADR-NNN: [Title]
Status: Accepted | Superseded | Proposed
Date: YYYY-MM-DD

Context
[Forces at play – technical, regulatory, project-scope. Value-neutral.]

Decision
We will [active voice statement].

Alternatives Considered

Alternative	Reason Rejected
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...	...

Evidence
[Citable sources with evidential grade: verified / estimated / speculative]

Consequences

- + [Positive consequence]
- [Negative consequence / acknowledged limitation]

ADR-001: Qwen3-4B as primary SLM (fully worked)

Status: Accepted · **Date:** 2026-03-15

Context. The Edge Negotiator requires a small language model that (a) runs on a developer laptop GPU with ≤8 GB VRAM, (b) carries a licence permitting use in transportation infrastructure, (c) supports both extended reasoning and low-latency single-token output within a single checkpoint. Three candidates were evaluated: Qwen3-4B (4.0B parameters, Apache 2.0), Phi-4-mini-instruct (3.8B parameters, MIT), and Llama 3.2 3B. Traffic-R1 Public 0.1 serves as a benchmark anchor but is not a deployment candidate.

Decision. We will use Qwen3-4B as the primary SLM, with Phi-4-mini as a comparison model and Traffic-R1 as a benchmark anchor.

Alternatives Considered

Alternative	Reason Rejected
Llama 3.2 3B	Meta's Acceptable Use Policy §2(d) explicitly prohibits "operation of critical infrastructure, transportation technologies, or heavy machinery" (GitHub +2) (https://www.llama.com/llama3_2/use-policy/ [archived: web.archive.org/web/2024/https://www.llama.com/llama3_2/use-policy/]). Meta's FAQ clarifies that critical infrastructure includes highways and traffic systems. (Llama) Even in simulation, developing algorithms intended for eventual traffic deployment would breach the AUP.
Phi-4-mini only	MIT-licensed and strong on reasoning benchmarks (https://arxiv.org/abs/2503.01743). However, Phi-4-mini lacks native dual thinking/non-thinking mode switching. The Edge Negotiator requires both a hot-path (non-thinking, single-token phase ID) and an audit-path (thinking, CoT justification). Qwen3's (enable_thinking) parameter provides this within a single checkpoint. (Hugging Face) Phi-4-mini is retained as a comparison model.
Traffic-R1	3B Qwen-based model fine-tuned for traffic signal control (https://arxiv.org/abs/2508.02344). Deployment claim of "55,000 drivers daily" is self-reported with no independent verification (Hugging Face) (see §2F, Evidential Standards). Released checkpoint is explicitly described as an "earlier checkpoint" omitting commercial training data. (Hugging Face) Retained as benchmark anchor only.

Evidence.

- Qwen3-4B model card confirms Apache 2.0 licence and dual-mode architecture (Hugging Face) (<https://huggingface.co/Qwen/Qwen3-4B> [archived: web.archive.org/web/2025/https://huggingface.co/Qwen/Qwen3-4B]). Qwen3 technical report: <https://arxiv.org/abs/2505.09388>.
- Phi-4-mini technical report: <https://arxiv.org/abs/2503.01743>.
- Llama 3.2 AUP confirmed at source URL above; Meta FAQ confirms "critical infrastructure" includes traffic systems (<https://www.llama.com/faq/> [archived: web.archive.org/web/2024/https://www.llama.com/faq/]). (Llama)

Consequences.

- Apache 2.0 licence removes all deployment-scope restrictions.
- Native dual-mode eliminates the need for two separate model checkpoints.
- 4.0B parameter count (Hugging Face) fits within laptop VRAM budget. – Qwen3-4B is a general-purpose model, not fine-tuned for traffic. Performance gap versus Traffic-R1 on

domain-specific benchmarks is expected and must be characterised empirically. – Post-hoc CoT from thinking mode is structurally unfaithful (see ADR-005, §2F).

ADR-002: Besu QBFT primary ledger, Tessera as empirical comparator (fully worked)

Status: Accepted · **Date:** 2026-03-18

Context. The Edge Negotiator requires a tamper-evident audit trail for SLM-generated traffic signal decisions. Two architectures were evaluated: a permissioned blockchain (Hyperledger Besu with QBFT consensus) and an append-only Merkle transparency log (Tessera, successor to Google Trillian). This decision is the most architecturally substantive in the project and directly supports the first pre-committed empirical contribution: quantifying cost-benefit trade-offs between blockchain audit and Merkle logging for AI decision provenance.

Decision. We will implement Besu QBFT as the primary audit backend (single Docker container, `--network=dev`, web3.py client, on-chain SHA-256 hash with off-chain 2 KB payload) and Tessera as a second backend. Both are committed implementations enabling head-to-head empirical comparison.

Alternatives Considered

Alternative	Reason Rejected as Sole Backend
Tessera only	Append-only Merkle log provides cryptographic integrity (GitHub) via inclusion and consistency proofs (GitHub) (https://github.com/transparency-dev/tessera [archived: web.archive.org/web/2025/https://github.com/transparency-dev/tessera]). However, Tessera is a single-operator log — it relies on external monitors to detect tampering. In the Edge Negotiator threat model, a compromised single operator can rewrite the log undetected if no independent monitor exists. Besu QBFT's multi-validator consensus ($f < n/3$ Byzantine tolerance) provides stronger tamper resistance even in its minimal 4-validator configuration. (Hyperledger) Tessera is retained as comparator to quantify the cost of this additional protection.
Public blockchain (Ethereum L1/L2)	Gas costs, latency (~12s block time), and dependency on external infrastructure make public chains unsuitable for real-time traffic signal audit.
SQLite + HMAC	Minimal overhead but no cryptographic integrity chain; trivially tamperable by the system operator.
IPFS + content addressing	Content-addressable storage provides integrity but no ordering guarantee or consensus; does not establish a tamper-evident timeline.

Evidence.

- Besu QBFT documentation confirms $\geq 2/3$ supermajority commit, immediate finality, minimum 4 validators for BFT (Hyperledger) (<https://besu.hyperledger.org/private-networks/how-to/configure/consensus/qbft> [archived: web.archive.org/web/2025/https://besu.hyperledger.org/private-networks/how-to/configure/consensus/qbft]).
- Tessera alpha release blog: <https://blog.transparency.dev/announcing-the-alpha-release-of-trillian-tessera> [archived: web.archive.org/web/2025/https://blog.transparency.dev/announcing-the-alpha-release-of-trillian-tessera].
- Certificate Transparency Merkle tree design using SHA-256 per RFC 6962 (GitHub) (<https://google.github.io/trillian/docs/TransparentLogging.html> [archived: web.archive.org/web/2024/https://google.github.io/trillian/docs/TransparentLogging.html])

Threat model cross-reference. The critical distinction is **single-operator versus multi-validator trust surface**. Tessera's threat model assumes an honest-but-curious operator checked by external monitors — appropriate for certificate transparency where monitors are Google, browser vendors, and certificate authorities. In a traffic signal context, no equivalent monitor ecosystem exists. Besu QBFT's consensus mechanism provides Byzantine fault tolerance intrinsically, (Hyperledger) shifting the threat from "operator must be monitored" to "fewer than one-third of validators must be compromised." The dissertation's empirical contribution quantifies the latency and storage cost of this stronger guarantee.

Consequences.

- Two-backend design directly produces the first empirical contribution (cost-benefit quantification).
 - Besu QBFT provides stronger tamper-resistance properties for safety-critical audit.
 - web3.py integration uses standard Ethereum JSON-RPC API. (GitHub) – Single Docker container in dev mode has no actual multi-validator consensus — this is a simulation constraint documented in ADR-003. – Besu has higher latency and storage overhead than Tessera; the dissertation quantifies this trade-off.
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ADR-003: Laptop simulation as hardware framing (fully worked)

Status: Accepted · **Date:** 2026-02-20

Context. The Edge Negotiator architecture targets edge deployment on intersection-local compute. However, procurement of edge hardware (NVIDIA Jetson, Raspberry Pi clusters) is outside dissertation scope, budget, and timeline. Both supervisors — Dr Akin Delibasi (UCL, academic) and Lee Stott (Microsoft, industry / UCL Honorary Associate Professor) — formally agreed that laptop-based SUMO simulation is sufficient for the MSc contribution.

Decision. We will evaluate the complete system on a single developer laptop using SUMO simulation via TraCI. Edge deployment is explicitly designated as future work.

Evidence.

- Supervisor approval recorded in supervision meeting minutes (2026-02-20). This is load-bearing evidence: the hardware scope was agreed before architectural design began.
- SUMO is the standard evaluation platform for traffic signal control research, used by Traffic-R1 (<https://arxiv.org/abs/2508.02344>), LLMLight (<https://dl.acm.org/doi/10.1145/3690624.3709379>), and the broader TSC literature. SUMO documentation: <https://eclipse.dev/sumo/> [archived: web.archive.org/web/2025/https://eclipse.dev/sumo/].

Consequences.

- Eliminates hardware procurement risk from the critical path.
 - SUMO provides reproducible, deterministic evaluation matching community practice. – All latency measurements are simulation-context only and cannot be directly extrapolated to edge hardware. The scope statement (§5) makes this explicit. – "Decentralised" architecture runs on a single machine — see §5 for pre-emptive viva framing.
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Skeletal ADRs (remaining four decisions)

ADR-004: Simplex safety layer with Z3 + MaxPressure fallback. *Context:* Safety-critical traffic control requires a formally verified fallback. *Decision:* Adopt Sha's Simplex architecture ([Semantic Scholar](#)) (<https://doi.org/10.1109/MS.2001.936213>) — SLM as high-performance controller, MaxPressure (Varaiya, 2013; [ResearchGate](#)) (<https://www.sciencedirect.com/science/article/abs/pii/S0968090X13001782>) as high-assurance controller, Z3 SMT solver ([Wikipedia](#)) (<https://github.com/Z3Prover/z3>) ([Z3prover](#)) [archived: web.archive.org/web/2025/https://github.com/Z3Prover/z3]) for safety-envelope verification. *Alternatives rejected:* Runtime Verification (overhead), pure rule-based (no learning). *Consequence:* MaxPressure is provably throughput-optimal ([ScienceDirect](#)) but suboptimal for individual intersection delay.

ADR-005: Hot-path single-token phase ID with CFG-constrained JSON. *Context:* Traffic signal decisions must complete within one simulation step. *Decision:* Constrain SLM output to single-token phase ID using XGrammar (<https://github.com/mlc-ai/xgrammar>) ([GitHub](#)) [archived: web.archive.org/web/2025/https://github.com/mlc-ai/xgrammar] or Outlines (<https://github.com/dottxt-ai/outlines>) ([GitHub](#)) [archived: web.archive.org/web/2025/https://github.com/dottxt-ai/outlines] CFG-constrained decoding, wrapped in structured JSON. *Alternatives rejected:* Free-text parsing (unreliable), multiple-token sequences (latency). *Consequence:* 100% structural correctness guaranteed; ([PyPI](#)) semantic correctness is the SLM's responsibility.

ADR-006: Audit-path CoT as post-hoc justification. *Context:* Regulatory transparency (EU AI Act Art. 13) requires explainable outputs. CoT is structurally post-hoc in a decide-first-justify-after architecture. *Decision:* Log CoT inside structured JSON with (`trace_type: "post_hoc_justification"`). Use prefix caching to amortise latency. *Evidence:* Turpin et al. (2023) show CoT can be systematically unfaithful ([Semantic Scholar](#)) ([Hugging Face](#)) (<https://arxiv.org/abs/2305.04388>); Lanham et al. (2023) show faithfulness decreases with model capability ([arXiv](#)) ([Hugging Face](#)) (<https://arxiv.org/abs/2307.13702>); Chen et al. (2025) confirm

reasoning models verbalise hint use <20% of the time (arXiv) (<https://arxiv.org/abs/2505.05410>).

Consequence: CoT is a monitorability signal, not a faithful causal record.

ADR-007: Inter-intersection communication protocol. *Context:* Intersections must share phase state and queue data. *Decision:* [TBD — recommend MQTT for implementation simplicity with Zenoh discussion]. MQTT requires minimal setup (paho-mqtt + Mosquitto Docker container), and broker latency is negligible on localhost simulation. *Alternatives rejected:* DDS (excessive complexity for MSc scope; multicast complications), Raw UDP (not credible for architecture claims). Zenoh is the strongest candidate for future edge deployment due to broker-optional topology and 5-byte wire overhead (Zenoh) (<https://zenoh.io/> [archived: web.archive.org/web/2025/https://zenoh.io/]). Comparative data from Zhang et al. (2024): <https://arxiv.org/abs/2309.07496>. *Consequence:* MQTT is a simulation convenience; the dissertation must not claim real-time suitability.

(B) Threat model: STRIDE × NIST AI RMF hybrid

Method selection rationale. Pure STRIDE (Shostack, 2014, ISBN 978-1-118-80999-0; Microsoft SDL: <https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-threats> [archived: web.archive.org/web/2025/https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-threats]) under-models AI-specific risks — hallucination, post-hoc rationalisation, and adversarial inputs to perception have no natural home in Spoofing/Tampering/Repudiation/Information Disclosure/DoS/Elevation of Privilege. (Wikipedia) Pure NIST AI RMF (NIST AI 100-1: (NIST) <https://doi.org/10.6028/NIST.AI.100-1>; (NIST) Generative AI Profile NIST AI 600-1: (Modelop) <https://doi.org/10.6028/NIST.AI.600-1>) under-models system-level attack vectors — network compromise, validator key theft, and communication channel tampering fall outside its AI-centric risk taxonomy. The hybrid maps STRIDE to infrastructure attack surface and NIST AI RMF to AI pipeline risks, with an explicit interface table for shared concerns.

Rejected alternatives:

Framework	Rejection Rationale
LINDDUN (https://linddun.org/ [archived: web.archive.org/web/2025/https://linddun.org/])	Privacy-only; does not cover integ or AI-specific threats. Useful supp narrow as primary method.
PASTA (UcedaVélez & Morana, 2015)	Seven-stage enterprise process re functional teams and threat intelli Disproportionate to MSc scope.
OCTAVE (https://www.sei.cmu.edu/library/introduction-to-the-octave-approach/ [archived: web.archive.org/web/2025/https://www.sei.cmu.edu/library/introduction-to-the-octave-approach/])	Organisational risk assessment re level staff workshops. Not architec
MITRE ATT&CK (https://attack.mitre.org/ [archived: web.archive.org/web/2025/https://attack.mitre.org/])	Post-exploitation TTP knowledge detection/red-teaming, not design modelling. MITRE ATLAS (https:// [archived: web.archive.org/web/2025/https:// ; covers adversarial ML but is likew catalogue, not a modelling method cited as supplementary references

The Shevchenko (2018) SEI comparison of 12 threat modelling methods (<https://www.sei.cmu.edu/blog/threat-modeling-12-available-methods/> [archived: web.archive.org/web/2025/https://www.sei.cmu.edu/blog/threat-modeling-12-available-methods/]) supports the position that STRIDE is the most appropriate design-time, architecture-focused method for individual researchers.

Interface table: STRIDE × NIST AI RMF shared concerns

Shared Concern	STRIDE Category	NIST AI RMF Category	Edge Negotiator Manifestation
Adversarial inputs	Tampering	Confabulation, Information Security	Ghost vehicles injected into SLM perception via spoofed sensor data or prompt injection via TraCI metadata
Output unreliability	Repudiation	Confabulation, Human-AI Configuration	SLM hallucinated phase selection; post-hoc CoT rationalising incorrect decision
Log tampering	Tampering, Repudiation	Value Chain Integration	Compromised audit log — single-operator (Tessera) vs multi-validator (Besu) threat surfaces
Communication compromise	Spoofing, Information Disclosure	Information Security	Spoofed inter-intersection messages causing coordination failure
Key compromise	Spoofing, Elevation of Privilege	Information Security	Insider theft of SLM-decision attestation keys enabling forged decision records

First page of threat model (five entries):

TM-001: Adversarial inputs to SLM perception. Ghost vehicles or prompt injection via TraCI metadata could cause the SLM to select incorrect signal phases. STRIDE: Tampering. NIST AI RMF: Confabulation (model acts on fabricated context), Information Security. Mitigation: Z3 safety-envelope verification rejects phases that violate flow-conservation constraints; MaxPressure fallback activates on safety violation.

TM-002: SLM hallucination / post-hoc rationalisation. The SLM may select a phase that no correct reasoning process would produce, then generate a plausible CoT justification. This is an *internal* threat — the model is not adversarial but structurally unreliable. Cross-reference: CoT is treated as a behavioural monitoring signal for anomaly detection, not as a faithful explanation of model computation (ADR-006, §2F). STRIDE: Repudiation (system cannot reliably account for its own decisions). NIST AI RMF: Confabulation.

TM-003: Compromised audit log. Single-operator Tessera log can be rewritten if the operator is compromised and no external monitor exists. Besu QBFT requires compromising $\geq 1/3$ validators. [Hyperledger](#) Cross-reference: ADR-002. STRIDE: Tampering, Repudiation. NIST AI RMF: Value Chain Integration.

TM-004: Compromised inter-intersection communication. Spoofed phase-state or queue-length messages between intersections could cause network-wide signal miscoordination. STRIDE: Spoofing, Information Disclosure. NIST AI RMF: Information Security. Mitigation: message authentication via HMAC; Simplex fallback to independent MaxPressure on authentication failure.

TM-005: Insider key compromise on SLM-decision attestation. An insider with access to the attestation private key can forge decision records indistinguishable from genuine SLM outputs. STRIDE: Spoofing, Elevation of Privilege. NIST AI RMF: Information Security. Mitigation: HSM key storage in production (out of dissertation scope); in simulation, key rotation and access logging.

(C) Requirements engineering from five codes of conduct

Requirements are derived from five codes of conduct — **algorithmic transparency, fail-safe override, data sovereignty, equitable service, adversarial robustness** — each mapped into four requirement classes using Value-Sensitive Design's tripartite methodology (Vt +2) (Friedman & Hendry, 2019; (MIT Press) <https://doi.org/10.7551/mitpress/7585.001.0001>) as the ethical analysis method. The compliance mapping draws on the EU AI Act (Regulation 2024/1689:

(European Commission) <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>), particularly Annex III §2 classifying AI in road traffic management as high-risk; GDPR Articles 5 and 22 (GDPR Info) (GDPR Info) (<https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>); the BCS Code of Conduct (BCS) (<https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/> [archived: web.archive.org/web/2025/https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/]); and the ACM Code of Ethics (Association for Computing ...) §§1.2, 1.4, 2.5, 2.9, 3.7 (<https://www.acm.org/code-of-ethics>).

Traceability structure (one row shown; full matrix in §4b):

Code	Requirement Class	Requirement	Acceptance Criterion	SUMO Test
Algorithmic Transparency	Functional	Every phase decision produces a logged CoT trace with <code>trace_type</code> label	100% of decisions have corresponding log entries with valid JSON schema	<code>test_audit_completeness</code> : run 1000-step simulation, assert log count = decision count
Algorithmic Transparency	Non-functional	Audit-path latency $\leq$ 500ms per decision	P99 CoT generation latency measured over 10,000 decisions	<code>test_audit_latency</code> : measure and assert P99 < 500ms
Algorithmic Transparency	Ethical (VSD)	Logged reasoning is accessible to non-expert stakeholders	Readability score of generated CoT $\geq$ Flesch-Kincaid grade 10	<code>test_cot_readability</code> : sample 100 CoT outputs, compute FK score
Algorithmic Transparency	Compliance	EU AI Act Art. 12 (record-keeping), Art. 13 (transparency)	System maintains automatic logs enabling traceability per Art. 12 requirements	<code>test_regulatory_logging</code> : verify log contains timestamp, input state, output phase, CoT, model version

The remaining four codes (fail-safe override, data sovereignty, equitable service, adversarial robustness) follow the same four-class decomposition. The full traceability matrix template is in §4(b).

(D) Trade-off analysis framework

Four methods were evaluated for architectural trade-off analysis:

Pugh matrix (Pugh, 1991, ISBN 978-0-201-41639-8): qualitative +/0/– scoring against a datum. (Gaudisite) (Wikipedia) Simple but lacks weighting granularity. **AHP** (Saaty, 1980, ISBN 978-0-07-054371-3): pairwise comparison with consistency checking via eigenvectors. (Scribd) Rigorous but requires subjective 1–9 scale judgments that are difficult to defend at viva without domain expert panels. **ATAM** (Kazman et al., CMU/SEI-2000-TR-004: <https://www.sei.cmu.edu/library/atam-method-for-architecture-evaluation/> [archived: web.archive.org/web/2025/https://www.sei.cmu.edu/library/atam-method-for-architecture-evaluation/]): identifies sensitivity and tradeoff points via utility trees and quality-attribute scenarios. (Software Engineering Instit...) Designed for multi-stakeholder workshops — disproportionate for individual MSc work. **MCDA** (Belton & Stewart, 2002; (Amazon) <https://doi.org/10.1007/978-1-4615-1495-4>): umbrella framework subsuming AHP, ELECTRE, PROMETHEE.

Recommendation: weighted Pugh matrix. The Pugh matrix is adopted as the primary method because it is transparent, reproducible, and does not require the subjective pairwise comparisons of AHP whose consistency ratio is difficult to defend without a panel. Criteria weights are derived from the requirements traceability matrix (§2C), making them traceable rather than arbitrary. ATAM's sensitivity/tradeoff point vocabulary is adopted as analytical language but its full workshop process is not.

Worked example: Besu QBFT vs Tessera (datum = Tessera)

Criterion (from requirements)	Weight	Tessera (datum)	Besu QBFT
Tamper resistance (multi-validator vs single-operator)	5	S	+
Write latency	3	S	–
Storage overhead per decision	2	S	–
Cryptographic integrity proof (inclusion/consistency)	4	S	S
Operational complexity (Docker containers)	3	S	–
Byzantine fault tolerance (intrinsic)	5	S	+
Independent monitor requirement	4	S	+
Weighted score		0	+6

This produces the defensible conclusion: **Besu QBFT is primary because its threat-model advantages (+10 on security-weighted criteria) outweigh its operational costs (–8 on latency/storage/complexity criteria).** Tessera is retained as the comparator to empirically

quantify the magnitude of these costs — the weighted Pugh matrix justifies the research question, not the answer.

(E) Evaluation framework for sociotechnical systems

Two-track evaluation design:

Track 1 — Technical. SUMO benchmarks on a 4×4 grid network (standard in TSC literature) measuring average waiting time, queue length, throughput, and number of stops. Ablation studies isolate contributions of each component (SLM vs MaxPressure baseline, with/without inter-intersection communication, with/without safety layer). Statistical tests: paired t-tests or Wilcoxon signed-rank across 30 random seeds, with Bonferroni correction for multiple comparisons.

Track 2 — Sociotechnical. Stakeholder analysis using VSD's conceptual investigation (Policy Review) (direct stakeholders: traffic engineers, road users; indirect: residents, emergency services). (Wikipedia) Value mapping from the five codes of conduct. (Wikipedia) Speculative design probes using Voros's futures cone (The Voroscope) (The Voroscope) (<https://doi.org/10.1108/14636680310698379>; Dunne & Raby, 2013, ISBN 978-0-262-01984-2): a **preferable 2035** scenario (equitable, transparent AI traffic management with cryptographic audit) and a **possible 2032** scenario (premature deployment without adequate safety verification).

Structural lessons from dissertations integrating both tracks:

Three dissertations were identified that rigorously integrate technical and sociotechnical evaluation. Davis (2009, University of Washington; <https://cs.whitman.edu/~davisj/pubs/davisjan-dissertation.pdf>) applied VSD's tripartite methodology to the UrbanSim transportation simulation, alternating between technical indicator development and value-focused stakeholder evaluation (Whitman) — the **two-track results chapter organisation** transfers directly. Wolters (TU Delft; <https://repository.tudelft.nl/islandora/object/uuid:c68f5beb-5e3b-44d0-995f-90dd8fb9085a>) used Design Science Research to develop a sociotechnical specification guide for ML systems, treating ML systems as "technical artefacts, human agents, and institutions" (TU Delft Repository) — the **stakeholder-mapped requirements traceability** pattern transfers. A third TU Delft dissertation (<https://repository.tudelft.nl/islandora/object/uuid:3683d5a1-1720-40ac-9781-6f356c7e1b6e>) proposed social readiness levels alongside Technology Readiness Levels — the **parallel readiness assessment** framing transfers.

What transfers: two-track results chapter organisation, stakeholder-mapped requirements traceability, parallel technical/social readiness assessment. **What does not transfer:** ethnographic fieldwork, longitudinal user studies, participatory design workshops. The

transferability argument protects the methodological choice; field-of-origin does not invalidate structural borrowing.

(F) Evidential standards

Evidence hierarchy (three grades):

Grade	Definition	Linguistic Markers	Source Treatment
Verified	Independently reproducible by the dissertation author	"we measure X", "our experiments show X", "we observe X"	Reproducible claims from the dissertation's own SUMO experiments
Estimated	Derived from credible external sources with stated assumptions	"we estimate X from Y", "according to [source], X", "based on [method], we infer X"	Peer-reviewed claims (with DOI), reproducible third-party benchmarks
Speculative	Extrapolated beyond available evidence	"we speculate that X", "if X holds, then Y would follow", "we hypothesise X"	Vendor claims without independent verification, single-study findings, architectural projections beyond tested conditions

Source treatment rules. Peer-reviewed publications with DOIs are treated as estimated-grade evidence unless independently reproduced. Vendor claims (model cards, blog posts, marketing materials) are treated as speculative-grade evidence unless corroborated. Reproducible claims from the dissertation's own experiments are the only verified-grade evidence.

Worked example 1: Traffic-R1 "55,000 drivers daily" claim. Traffic-R1 (<https://arxiv.org/abs/2508.02344>) reports that "the model now manages signals for more than 55,000 drivers daily, shortening average queues by over 5%." This is a self-reported deployment claim. No independent municipal authority or third-party researcher has confirmed the deployment. The Hugging Face model card notes the released checkpoint omits commercial training data (<https://huggingface.co/Season998/Traffic-R1> [archived: web.archive.org/web/2025/https://huggingface.co/Season998/Traffic-R1]). The deployment location is unspecified.

Suggested wording: *"Zou et al. (2025) report a deployment serving 55,000 drivers daily with a 5% queue reduction; however, this claim is self-reported and has not been independently verified*

as of April 2026. We treat Traffic-R1's simulation benchmarks on CityFlow datasets as estimated-grade evidence and the deployment claim as speculative-grade evidence."

Worked example 2: CoT logging claim. The Edge Negotiator logs CoT generated *after* the phase decision has been made. Turpin et al. (2023) demonstrate that CoT explanations can be systematically biased by features the model fails to mention (<https://arxiv.org/abs/2305.04388>). Lanham et al. (2023) show that larger, more capable models produce *less* faithful reasoning on most tasks (<https://arxiv.org/abs/2307.13702>). Chen et al. (2025) find that reasoning models verbalise hint usage less than 20% of the time (<https://arxiv.org/abs/2505.05410>). Arcuschin et al. (2025) document post-hoc rationalisation rates of 7–13% in production models on realistic prompts (<https://arxiv.org/abs/2503.08679>).

Suggested wording: *"The logged reasoning trace is treated as a behavioural signal for anomaly detection, not as a faithful record of the model's internal computation. The `trace_type: 'post_hoc_justification'` label encodes this epistemic status in the data schema. We measure distributional properties of CoT outputs (length, vocabulary, consistency with decision) as monitorability signals; we do not claim these traces explain why the model selected a given phase."*

Label discipline: In code, every CoT log entry carries `trace_type: "post_hoc_justification"`. In prose, every reference to CoT uses "behavioural signal" or "monitorability signal" rather than "explanation" or "reasoning trace."

§3 Proposed methodology chapter skeleton

This is the primary deliverable. Each section includes a target word count and a one-sentence description. Total target: ~3,000 words within a ~15,000 word dissertation.

§	Heading	Words	Description
3.1	Chapter scope and relationship to research questions	150	States that this chapter documents architectural decisions, derives requirements, and declares evidential standards, linking each to specific research questions.
3.2	Architecture Decision Records	600	Presents the ADR template and all seven ADRs (three fully worked: Qwen3-4B, Besu/Tessera, hardware framing; four skeletal: safety layer, hot-path, audit-path, comms protocol).
3.3	Threat model: STRIDE × NIST AI RMF hybrid	450	Justifies the hybrid method by rejecting LINDDUN, PASTA, OCTAVE, and MITRE ATT&CK; presents the interface table; documents the first five threat entries spanning adversarial inputs, hallucination, audit compromise, communication spoofing, and key compromise.
3.4	Requirements derivation from codes of conduct	400	Maps five codes of conduct (algorithmic transparency, fail-safe override, data sovereignty, equitable service, adversarial robustness) through four requirement classes (functional, non-functional, ethical via VSD, compliance) to acceptance criteria and SUMO tests.
3.5	Trade-off analysis: method selection and worked example	350	Compares Pugh, AHP, ATAM, MCDA; selects weighted Pugh matrix; works through Besu vs Tessera scoring to produce the "Besu primary, Tessera comparator" conclusion.
3.6	Evaluation framework: technical and sociotechnical tracks	400	Defines the two-track evaluation design — SUMO benchmarks with statistical rigour on Track 1, VSD stakeholder analysis and speculative design probes on Track 2 — citing structural lessons from Davis (2009), Wolters (TU Delft), and parallel readiness assessment literature.
3.7	Evidential standards and epistemic discipline	350	Declares the three-grade evidence hierarchy (verified/estimated/speculative), establishes linguistic markers, and presents the two worked examples (Traffic-R1 deployment claim, CoT faithfulness).
3.8	Scope limitations and pre-emptive viva framing	250	Contains the honest scope statement (§5 below), explicitly addressing laptop simulation vs edge deployment, single-machine decentralisation, blockchain choice constraints, and structural post-hoc CoT.
Total		~2,950	

§4 Document templates

(a) Architecture Decision Record template

markdown

ADR-[NNN]: [Short noun phrase title]

****Status:**** [Proposed | Accepted | Deprecated | Superseded by ADR-XXX]

****Date:**** [YYYY-MM-DD]

****Deciders:**** [Names and roles]

Context

[Describe the forces at play – technical, regulatory, project-scope, resource constraints. Write in value-neutral language. 2-4 sentences.]

Decision

We will [active voice statement of the decision].

Alternatives Considered

Alternative	Reason Rejected
[Option A]	[Specific technical/regulatory/scope reason]
[Option B]	[Specific reason]

Evidence

Claim	Source	Evidential Grade
[Claim 1]	[URL / DOI]	[Verified / Estimated / Speculative]
[Claim 2]	[URL / DOI]	[Grade]

Consequences

- + [Positive consequence 1]
- + [Positive consequence 2]
- [Negative consequence or acknowledged limitation 1]
- [Negative consequence 2]

Cross-references

- Threat model entry: [TM-NNN]
- Requirement: [REQ-NNN]
- Related ADR: [ADR-NNN]

(b) Requirements traceability matrix template

markdown

ID	Code of Conduct	Req. Class	Requirement Statement	Acceptance Criterion
REQ-001	Algorithmic Transparency	Functional	[What the system does]	[Measure]
REQ-002	Algorithmic Transparency	Non-functional	[Performance/latency target]	
REQ-003	Algorithmic Transparency	Ethical (VSD)	[Value identified via VSD con]	
REQ-004	Algorithmic Transparency	Compliance	[EU AI Act Art. / GDPR Art. / BC]	
REQ-005	Fail-Safe Override	Functional	[...] [...] [...] [...] [...]	

Populate 4 rows per code of conduct × 5 codes = 20 minimum requirements.

(c) Trade-off matrix template (weighted Pugh)

Criterion	Weight (1-5)	Source of Weight	Datum: [Option A]	[Option B]	[Op
[Criterion 1 from requirements]	[W]	[REQ-NNN]	S	[+/S/-]	[+/S/-]
[Criterion 2]	[W]	[REQ-NNN]	S	[+/S/-]	[+/S/-]
Weighted + total		**0**	**[Σ]**	**[Σ]**	
Weighted - total		**0**	**[Σ]**	**[Σ]**	
Net weighted score		**0**	**[Σ]**	**[Σ]**	
Decision: [Option X] is selected as primary because [1-sentence rationale linking to weighted score and threat model]. [Option Y] is retained as [comparator/fallback] because [rationale].					

(d) Threat model entry template

--

TM-[NNN]: [Threat title]

****STRIDE Category:**** [Spoofing | Tampering | Repudiation | Info Disclosure | DoS | E
****NIST AI RMF Category:**** [Confabulation | Information Security | Human-AI Configura
****Interface concern:**** [If this threat spans both frameworks, state the shared conce

Description

[2-3 sentences: who is the threat actor, what is the attack vector,
what is the impact on the Edge Negotiator.]

Affected Components

- [Component 1, e.g., SLM inference pipeline]
- [Component 2, e.g., Besu audit log]

Mitigation

Mitigation	Implementation Status	Linked ADR
[Mitigation 1]	[Implemented Designed Future work]	ADR-NNN

Residual Risk

[1-2 sentences: what risk remains after mitigation, and why it is
accepted within dissertation scope.]

Evidential Grade of Threat Assessment

[Verified: threat demonstrated in own experiments |
Estimated: threat documented in literature [cite] |
Speculative: threat hypothesised from architectural analysis]

(e) Evidential claim template with hedging language

markdown

CLAIM-[NNN]: [One-sentence claim]

Evidential Grade: [Verified | Estimated | Speculative]

Evidence

Source	Type	URL/DOI	Independence
[Source 1]	[Own experiment Peer-reviewed Vendor claim Self-reported]	[UR	

Suggested Prose (copy-edit to context)

If Verified:

"We measure [X] across [N] experimental runs (§[section]).
The observed value is $[Y \pm CI]$."

If Estimated:

"Based on [source]'s reported [metric] of [value], we estimate [X].
This estimate assumes [stated assumption]; if [assumption] does not hold,
[X] would [direction of change]."

If Speculative:

"We speculate that [X], extrapolating from [limited evidence].
This claim has not been independently verified and should be treated
as a hypothesis for future work rather than an established finding."

Anti-patterns (do NOT use)

- ✗ "X proves that..." (nothing in an MSc dissertation proves anything)
- ✗ "It is well-known that..." (appeal to authority without citation)
- ✗ "X shows that..." when source is a vendor blog (overstates evidence)
- ✗ Citing Traffic-R1's deployment numbers without "self-reported" qualifier
- ✗ Describing CoT as "the model's reasoning" without "post-hoc" qualifier

§5 Honest scope statement (draft clause, 240 words)

Scope limitations and pre-emptive framing. This dissertation evaluates the Edge Negotiator architecture entirely within SUMO simulation on a single developer laptop. This hardware framing was formally agreed with both supervisors before architectural design began (supervision minutes, 2026-02-20) and reflects a deliberate scope decision, not an oversight. Four limitations follow directly and are acknowledged here to pre-empt mischaracterisation at viva.

Laptop simulation vs edge deployment. All latency, throughput, and resource-utilisation measurements are simulation-context values. They cannot be directly extrapolated to edge hardware (e.g., NVIDIA Jetson Orin) without profiling on target silicon, which is designated as future work.

Single-machine decentralisation. Hyperledger Besu runs in dev mode on one Docker container. The QBFT consensus properties documented in ADR-002 — Byzantine fault tolerance requiring ≥ 4 validators — are architectural properties of the protocol, not empirical properties of this deployment. The Besu Quorum Developer Quickstart (4 validators via Docker Compose) could strengthen this claim within laptop resources and is identified as a stretch goal.

Blockchain choice constraints. The Besu-vs-Tessera comparison is bounded by single-machine performance. Network latency, validator geographic distribution, and cross-organisational governance are out of scope.

Structural post-hoc CoT. The decide-first-justify-after architecture means all logged Chain-of-Thought is structurally post-hoc. Per Turpin et al. (2023), Lanham et al. (2023), and Chen et al. (2025), such traces are treated throughout this dissertation as behavioural monitoring signals, not as faithful records of model computation. The `trace_type:` `"post_hoc_justification"` label enforces this epistemic commitment in the data schema itself.